

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.SC. DIGITAL FORENSICS AND INFORMATION SECURITY - Semester - II - Apr-2026

Subject Code: CTMSDFIS24 SII P2
Subject Name: Web Application Security
Time: 2.30 PM TO 5.30 PM
Total Marks: 100

Date: 27/04/26

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
(a)	Apply the concepts of cookies and sessions in a web application to demonstrate how user login state is maintained across multiple requests.	08
(b)	Analyze the importance of subdomain enumeration in web security assessment and examine how incomplete enumeration affects vulnerability discovery.	08
(c)	Apply web server fingerprinting techniques to identify the technology stack of a given web server and suggest possible tools used.	08
(d)	Analyze the communication process of TCP and HTTPS protocols.	08
Q.2	Attempt any three.	
(a)	What is a vulnerability in a web application and give any two examples. List different severity levels defined in CVSS and explain how they are useful.	08
(b)	Analyze the relationship between CVE and CWE and differentiate their roles in vulnerability identification and classification with examples.	08
(c)	Apply the STRIDE threat modeling framework to identify potential threats in a web application system.	08
(d)	Analyze how the DREAD model helps in prioritizing security risks and supports decision-making in threat evaluation.	08
Q.3	Attempt any three.	
(a)	What is web spidering? Differentiate between traditional spider and AJAX spider in ZAP tool.	08
(b)	Explain how the following attacks can be performed using the ffuf tool. 1. Directory & File Discovery 2. GET Parameter Fuzzing 3. API Endpoint Discovery	08

- (c) What are some best practices for password storage? Explain any three Multi-Factor Authentications used in modern applications with suitable examples. 08
- (d) A user connects to an unsecured public WiFi network. Explain any three possible attacks on the user online activities in this scenario with examples. 08

Q.4

Attempt any two.

- (a) Write a short note on the following vulnerabilities in web applications and give appropriate examples for each. 07
1. Injection
 2. Sensitive Data Exposure
 3. Insecure Deserialization
- (b) A web application allows users to view different pages using a URL parameter: **http://example.com/index.php?page=home**
Attacker modifies the URL to: **http://example.com/index.php?page=http://malicious-site.com/access**
The application loads and executes the external file, resulting in unauthorised access to the server.
Identify the type of attack launched on the web server? What vulnerabilities caused the web application to run malicious script? How do you patch the vulnerabilities to prevent these types of attacks? 07
- (c) Explain how SSRF attacks are launched on web applications? What are the impacts of SSRF attacks on back-end systems? 07

Q.5

Attempt any two.

- (a) List the key characteristics of Web Service Architecture and explain how it supports interoperability. 07
- (b) Explain Information Leakage via APIs with the help of any two examples. List different causes and countermeasures for Information Leakage via APIs. 07
- (c) Explain Content Management System (CMS) with a use case in the context of a university. 07

END OF PAPER